**148.** *(c)* $\because \quad \sin\alpha = \frac{\sqrt{3}}{2} \Rightarrow \sin\alpha = \sin 60°$

$\Rightarrow \quad \alpha = 60° \quad \left(\because \sin 60° = \frac{\sqrt{3}}{2}\right)$

Now, $\cos\beta = \frac{\sqrt{3}}{2} \Rightarrow \beta = 30° \quad \left(\because \cos 30° = \frac{\sqrt{3}}{2}\right)$

and $\tan\gamma = 1 \Rightarrow \gamma = 45° \quad (\because \tan 45° = 1)$

$\therefore \alpha + \beta + \gamma = 60° + 30° + 45° = 135°$

**149.** *(a)* $$\frac{(1+\sec\theta - \tan\theta)\cos\theta}{(1+\sec\theta+\tan\theta)(1-\sin\theta)} = \frac{\left(1+\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}\right)\cos\theta}{\left(1+\frac{1}{\cos\theta}+\frac{\sin\theta}{\cos\theta}\right)(1-\sin\theta)}$$

$$= \frac{\left(\frac{\cos\theta+1-\sin\theta}{\cos\theta}\right)\cos\theta}{\frac{(\cos\theta+1+\sin\theta)(1-\sin\theta)}{\cos\theta}}$$

$$= \frac{\cos\theta+1-\sin\theta}{\frac{\cos\theta+1+\sin\theta-\sin\theta\cos\theta-\sin\theta-\sin^2\theta}{\cos\theta}}$$

$$= \frac{\cos\theta+1-\sin\theta}{\frac{\cos\theta+1-\sin^2\theta-\sin\theta\cos\theta}{\cos\theta}} \quad (\because 1-\sin^2\theta = \cos^2\theta)$$

$$= \frac{\cos\theta+1-\sin\theta}{\frac{\cos\theta+\cos^2\theta-\sin\theta\cos\theta}{\cos\theta}} = \frac{\cos\theta+1-\sin\theta}{\frac{\cos\theta\,(\cos\theta+1-\sin\theta)}{\cos\theta}} = 1$$

**150.** *(c)* In right angled $\Delta ABC$, if $\angle C$ is 90°, then

$\angle A + \angle B = 180° - 90° = 90°$

Now, $\cos(A+B) + \sin(A+B) = \cos 90° + \sin 90° = 0 + 1 = 1$

**151.** *(a)* Only statement I is correct as $\tan\theta$ increases faster than $\sin\theta$ as $\theta$ increases while statement II is wrong as the value of $\sin\theta + \cos\theta$ is not always greater than 1. It may also be equal to 1.

**152.** *(a)* Since, value of $\cos\theta$ decreases, from 0° to 90° and at 45° it is equal to the value of $\sin\theta$.

Similarly, value of $\sin\theta$ increases from 0 to 90° and at 45° it is equal to the value of $\cos\theta$.

For $0° < \theta < 45°$, $\cos\theta > \sin\theta$

So, value of $\cos 25° - \sin 25°$ is always positive but less than 1.

**153.** *(b)* In right angled $\Delta ABC$, $\angle B = 90°$

$\angle C = 180° - (\angle B + \angle A) = 180° - 90° - \angle A = 90° - \angle A$

$\therefore \sin C = \sin(90° - A) = \cos A = 4/5$

**154.** *(c)* Since, $\alpha$ and $\beta$ are complementary angle.

$\therefore \alpha = 90° - \beta$

Now, $\sqrt{\cos\alpha\,\mathrm{cosec}\beta - \cos\alpha\sin\beta} = \sqrt{\frac{\cos\alpha}{\sin\beta} - \cos\alpha\sin\beta}$

$= \sqrt{\frac{\cos\alpha}{\cos(90°-\beta)} - \cos\alpha\cos(90°-\beta)}$

$= \sqrt{\frac{\cos\alpha}{\cos\alpha} - \cos\alpha\cdot\cos\alpha} = \sqrt{1-\cos^2\alpha} = \sqrt{\sin^2\alpha} = \sin\alpha$

**155.** *(d)* Given, $\sec\theta + \tan\theta = 2$ ...(i)

By trigonometric identity, $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1 \Rightarrow \sec\theta - \tan\theta = 1/2$ ...(ii)

On adding Eqs. (i) and (ii), we get

$\Rightarrow \quad 2\sec\theta = \frac{1}{2} + 2 \Rightarrow \sec\theta = \frac{5}{4}$

**156.** *(b)* $\mathrm{cosec}(75° + \theta) - \sec(15° - \theta) - \tan(55° + \theta) + \cot(35° - \theta)$

$= \mathrm{cosec}(75° + \theta) - \sec[90° - (75° + \theta)]$

$- \tan(55° + \theta) + \cot[90° - (55° + \theta)]$

$= \mathrm{cosec}(75° + \theta) - \mathrm{cosec}(75° + \theta)$

$- \tan(55° + \theta) + \tan(55° + \theta) = 0$

**157.** *(d)* $\sin 25° \sin 35° \sec 65° \sec 55°$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\cos 65°} \cdot \frac{1}{\cos 55°}$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\cos(90°-25°)} \cdot \frac{1}{\cos(90°-35°)}$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\sin 25°} \cdot \frac{1}{\sin 35°} = 1$

**158.** *(c)* Given, $2\cot\theta = 3 \Rightarrow \cot\theta = \frac{3}{2}$

$$\therefore \frac{2\cos\theta - \sin\theta}{2\cos\theta + \sin\theta} = \frac{2\cot\theta - 1}{2\cot\theta + 1} = \frac{2\times\frac{3}{2} - 1}{2\times\frac{3}{2} + 1} = \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2}$$

**159.** *(d)* $\sin^6\theta + \cos^6\theta = (\sin^2\theta)^3 + (\cos^2\theta)^3$

$= (\sin^2\theta + \cos^2\theta)(\sin^4\theta + \cos^4\theta - \sin^2\theta\cos^2\theta)$

$[\because a^3 + b^3 = (a+b)(a^2 + b^2 - ab)]$

$= (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta - \sin^2\theta\cos^2\theta$

$= (1 - 3\sin^2\theta\cos^2\theta) = 1 - 3\times\frac{1}{4} = 1 - \frac{3}{4} = \frac{1}{4}$

**160.** *(a)* Given, $\cos x + \sec x = 2$ ...(i)

$\Rightarrow \cos x + \frac{1}{\cos x} = 2 \Rightarrow \cos^2 x + 1 = 2\cos x$

$\Rightarrow \cos^2 x - 2\cos x + 1 = 0 \Rightarrow (\cos x - 1)^2 = 0 \Rightarrow \cos x = 1$

$\therefore \sec x = \frac{1}{\cos x} = 1$

$\therefore \cos^n x + \sec^n x = (1)^n + (1)^n = 1 + 1 = 2$

**161.** *(c)* Given, $\sin\theta + 2\cos\theta = 1$

On squaring both sides, we get $(\sin\theta + 2\cos\theta)^2 = 1$

$\Rightarrow \quad \sin^2\theta + 4\cos^2\theta + 4\sin\theta\cos\theta = 1$

$\Rightarrow \quad (1 - \cos^2\theta) + 4(1 - \sin^2\theta) + 4\sin\theta\cos\theta = 1$

$\Rightarrow \quad -(\cos^2\theta + 4\sin^2\theta) + 4\sin\theta\cos\theta = 1 - 5$

$\Rightarrow \quad \cos^2\theta + 4\sin^2\theta - 4\sin\theta\cos\theta = 4$

$\Rightarrow \quad (2\sin\theta - \cos\theta)^2 = 4 \Rightarrow 2\sin\theta - \cos\theta = 2$

**162.** *(b)* Given, $\tan 8\theta = \cot 2\theta$

$\Rightarrow \quad \tan 8\theta = \tan(90° - 2\theta) \Rightarrow 8\theta = 90° - 2\theta \Rightarrow \theta = 9°$

$\therefore \quad \tan 5\theta \Rightarrow \tan 45° = 1$

**163.** *(a)* $\because \sin(A + B) = 1 = \sin 90° \Rightarrow (A + B) = 90°$

$\Rightarrow A = 90° - B$

Now, $\cos(A - B) = \cos A \cos B + \sin A \sin B$

$= \cos(90° - B)\cos B + \sin(90° - B)\sin B$

$= \sin B\cos B + \cos B\sin B = 2\sin B\cos B = \sin 2B$

**164.** *(d)* Clock will make right angle at $(5n + 15)\times\frac{12}{11}$ min past $n$.

Here, $n = 3$

$\therefore (5\times 3 + 15)\times\frac{12}{11}$ min past 3 $= 30\times\frac{12}{11}$ min past 3

$= 32\frac{8}{11}$ min past 3 i.e. 3 h $32\frac{8}{11}$ min

**165.** *(c)* Given, $\tan\theta + \cot\theta = 2$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 2 \Rightarrow \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = 2$$

$$\Rightarrow \frac{1}{\sin\theta\cos\theta} = 2 \Rightarrow \sin\theta\cos\theta = \frac{1}{2} \quad \text{...(i)}$$

Now, $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$

$$= 1 + 2 \times \frac{1}{2} = 1 + 1 \quad \text{[from Eq. (i)]}$$

$$\Rightarrow (\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin\theta + \cos\theta = \sqrt{2}$$

**166.** *(a)* $\dfrac{\sec x}{\cot x + \tan x} = \dfrac{1/\cos x}{\dfrac{\cos x}{\sin x} + \dfrac{\sin x}{\cos x}}$

$$= \frac{1/\cos x}{\dfrac{\cos^2 x + \sin^2 x}{\sin x \cos x}} = \frac{1}{\cos x} \times \frac{\sin x\cos x}{1} = \sin x$$

**167.** *(b)* $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$

$$= (1 + \cot x - \operatorname{cosec} x)\left(1 + \frac{1}{\cot x} + \sec x\right)$$

$$= \frac{(1 + \cot x - \operatorname{cosec} x)(1 + \cot x + \operatorname{cosec} x)}{\cot x}$$

$$= \frac{(1 + \cot x)^2 - (\operatorname{cosec} x)^2}{\cot x}$$

$$= \frac{1^2 + \cot^2 x + 2\cot x - \operatorname{cosec}^2 x}{\cot x}$$

$$= \frac{1 + 2\cot x - (\operatorname{cosec}^2 x - \cot^2 x)}{\cot x} = \frac{1 + 2\cot x - 1}{\cot x} = 2$$

**168.** *(d)* $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$

$$= \left(\frac{1}{\sin x} - \sin x\right)\left(\frac{1}{\cos x} - \cos x\right)\left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}\right)$$

$$= \frac{(1 - \sin^2 x)(1 - \cos^2 x)(\sin^2 x + \cos^2 x)}{\sin x\cos x \cdot \sin x\cos x} = \frac{\cos^2 x\sin^2 x \times 1}{\sin^2 x\cos^2 x} = 1$$

**169.** *(b)* $\dfrac{\sin x - \cos x + 1}{\sin x + \cos x - 1} = \dfrac{(\sin x - \cos x) + 1}{(\sin x + \cos x) - 1} \times \dfrac{(\sin x + \cos x) + 1}{(\sin x + \cos x) + 1}$

$$= \frac{(\sin x - \cos x + 1)(\sin x + \cos x + 1)}{(\sin x + \cos x)^2 - 1}$$

$$[\because (a - b)(a + b) = a^2 - b^2]$$

$$= \frac{\sin^2 x + \sin x\cos x + \sin x - \cos x\sin x - \cos^2 x - \cos x + \sin x + \cos x + 1}{\sin^2 x + \cos^2 x + 2\sin x\cos x - 1}$$

$$= \frac{\sin^2 x + 2\sin x - \cos^2 x + 1}{1 + 2\sin x\cos x - 1}$$

$$= \frac{\sin^2 x + 2\sin x - (1 - \sin^2 x) + 1}{2\sin x\cos x} \quad [\because \cos^2 x = 1 - \sin^2 x]$$

$$= \frac{\sin^2 x + 2\sin x - 1 + \sin^2 x + 1}{2\sin x\cos x}$$

$$= \frac{2\sin^2 x + 2\sin x}{2\sin x\cos x} = \frac{2\sin x(\sin x + 1)}{2\sin x\cos x} = \frac{\sin x + 1}{\cos x}$$

**170.** *(b)* $(\sin^2 x - \cos^2 x)(1 - \sin^2 x\cos^2 x)$

$$= (\sin^2 x - \cos^2 x)[(\sin^2 x + \cos^2 x)^2 - \sin^2 x\cos^2 x]$$

$$(\because \sin^2 x + \cos^2 x = 1)$$

$$= (\sin^2 x - \cos^2 x)[(\sin^4 x + \cos^4 x + 2\sin^2 x\cos^2 x) - \sin^2 x\cos^2 x]$$

$$= (\sin^2 x - \cos^2 x)(\sin^4 x + \cos^4 x + \sin^2 x\cos^2 x)$$

$$= (\sin^2 x)^3 - (\cos^2 x)^3 \quad [\because (a - b)(a^2 + b^2 + ab) = a^3 - b^3]$$

$$= \sin^6 x - \cos^6 x$$

**171.** *(d)* $(\sin x \cdot \cos y + \cos x \cdot \sin y)(\sin x \cdot \cos y - \cos x \cdot \sin y)$

$$= \sin(x + y) \cdot \sin(x - y) = \sin^2 x - \sin^2 y$$

$$[\because \sin^2 A - \sin^2 B = \sin(A + B)\sin(A - B)]$$

**172.** *(d)* In $\Delta ABC$, $A + B + C = 180^\circ \Rightarrow A + B = 180^\circ - C$

$$\tan(A + B) = \tan(180^\circ - C) = -\tan C$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A\tan B} = -\tan C$$

$\tan A + \tan B + \tan C = \tan A\tan B\tan C$

On dividing both sides by $\tan A\tan B\tan C$

$$\frac{1}{\tan B\tan C} + \frac{1}{\tan A\tan C} + \frac{1}{\tan A\tan B} = 1$$

$$\Rightarrow \cot B\cot C + \cot A\cot C + \cot A\cot B = 1$$

**173.** *(a)* $\sin x + \operatorname{cosec} x = 2 \Rightarrow \sin x + \dfrac{1}{\sin x} = 2 \Rightarrow \sin^2 x + 1 = 2\sin x$

$$\Rightarrow \sin^2 x - 2\sin x + 1 = 0 \Rightarrow (\sin x - 1)^2 = 0 \Rightarrow \sin x = 1$$

$$\operatorname{cosec} x = \frac{1}{\sin x} = 1$$

$$\therefore \sin^9 x + \operatorname{cosec}^9 x = (1)^9 + (1)^9 = 1 + 1 = 2$$

**174.** *(b)* Given, $\sin x + \cos x = p$ ...(i)

and $\sin^3 x + \cos^3 x = q$ ...(ii)

On cubing Eq. (i) on both sides, we get

$\sin^3 x + \cos^3 x + 3\sin x\cos x(\sin x + \cos x) = p^3$

$\Rightarrow q + 3\sin x\cos x(p) = p^3$ [from Eqs. (i) and (ii)] ...(iii)

On squaring Eq. (i) on both sides, we get

$\sin^2 x + \cos^2 x + 2\sin x\cos x = p^2$

$$\Rightarrow \sin x\cos x = \frac{p^2 - 1}{2} \quad (\because \sin^2 x + \cos^2 x = 1)$$

On putting this value in Eq. (iii), we get $q + \dfrac{3(p^2 - 1)p}{2} = p^3$

$$\Rightarrow 2q + 3p^3 - 3p = 2p^3 \Rightarrow p^3 - 3p = -2q$$

**175.** *(d)* Since, 1 radian = 57°16′22″

So, sin 1° < sin 1 and cos 1° > cos 1

Hence, neither statement I nor II is correct.

**176.** *(b)* $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$

$$= \operatorname{cosec}^2(90^\circ - 23^\circ) + \sec^2(90^\circ - 33^\circ) - \cot^2 33^\circ - \tan^2 23^\circ$$

$$= \sec^2 23^\circ + \operatorname{cosec}^2 33^\circ - \cot^2 33^\circ - \tan^2 23^\circ$$

$$= 1 + \tan^2 23^\circ + 1 + \cot^2 33^\circ - \cot^2 33^\circ - \tan^2 23^\circ = 2$$

$$(\because 1 + \tan^2 q = \sec^2 q \text{ and } 1 + \cot^2 q = \operatorname{cosec}^2 q)$$

**177.** *(c)* Given, $\tan(A + B) = \sqrt{3} = \tan 60^\circ$

$\therefore A + B = 60^\circ$ ...(i)

and $\tan A = 1 \Rightarrow \tan A = \tan 45^\circ$

$\therefore A = 45^\circ$

From Eq. (i), $A + B = 60^\circ \Rightarrow 45^\circ + B = 60^\circ \Rightarrow B = 15^\circ$

Now, $\tan(A - B) = \tan(45^\circ - 15^\circ) = \tan 30^\circ = 1/\sqrt{3}$

Hence, the value of $\tan(A - B)$ is $1/\sqrt{3}$.

**178.** *(d)* Given, $\tan A + \cot A = 4$

On squaring both sides, we get

$$(\tan A + \cot A)^2 = (4)^2$$

$\Rightarrow \tan^2 A + \cot^2 A + 2\tan A \cot A = 16$

$\Rightarrow \quad \tan^2 A + \cot^2 A + 2 = 16$

$\Rightarrow \quad \tan^2 A + \cot^2 A = 14$

Again, squaring both sides, we get

$$(\tan^2 A + \cot^2 A)^2 = (14)^2$$

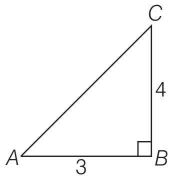
$\Rightarrow \quad \tan^4 A + \cot^4 A + 2\tan^2 A \cot^2 A = 196$

$\Rightarrow \quad \tan^4 A + \cot^4 A + 2 = 196$

$\Rightarrow \quad \tan^4 A + \cot^4 A = 194$

**179.** *(c)* In right angled $\Delta ABC$, $AB : BC = 3 : 4$

or $\dfrac{AB}{BC} = \dfrac{3}{4}$




Now, in $\Delta ABC$

$AC^2 = AB^2 + BC^2$

$= 3^2 + 4^2$

$= 9 + 16 = 25$

$\Rightarrow AC = 5 \Rightarrow \sin A = \dfrac{BC}{AC} = \dfrac{4}{5}$

$\sin B = 90° = \sin 90° = 1, \sin C = \dfrac{AB}{AC} = \dfrac{3}{5}$

Now, $\sin A + \sin B + \sin C = \dfrac{4}{5} + 1 + \dfrac{3}{5}$

$= \dfrac{4+5+3}{5} = \dfrac{12}{5}$

**180.** *(b)* Given, $\sin x + \cos x = C$

On squaring both sides, we get

$$(\sin x + \cos x)^2 = C^2$$

$\Rightarrow \quad \sin^2 x + \cos^2 x + 2\sin x \cos x = C^2$

$\Rightarrow \quad 1 + 2\sin x \cos x = C^2$

$\Rightarrow \quad \sin x \cos x = \dfrac{C^2 - 1}{2} \quad \ldots\text{(i)}$

Now, $\sin^6 x + \cos^6 x = (\sin^2 x)^3 + (\cos^2 x)^3$

$= (\sin^2 x + \cos^2 x)[\sin^4 x + \cos^4 x - \sin^2 x \cos^2 x]$

$[\because a^3 + b^3 = (a+b)(a^2 + b^2 - ab)]$

$= 1[(\sin^2 x + \cos^2 x)^2 - 3\sin^2 x \cos^2 x]$

$= (1 - 3\sin^2 x \cos^2 x)$

$\therefore \sin^6 x + \cos^6 x = 1 - 3\sin^2 x \cos^2 x$

$= 1 - 3\left(\dfrac{C^2 - 1}{2}\right)^2 \quad$ [from Eq. (i)]

$= 1 - 3\left(\dfrac{C^4 + 1 - 2C^2}{4}\right)$

$= \dfrac{4 - 3C^4 - 3 + 6C^2}{4} = \dfrac{1 + 6C^2 - 3C^4}{4}$

**181.** *(c)* Given, $\dfrac{3 - \tan^2 A}{1 - 3\tan^2 A} = k$

$\Rightarrow \quad 3 - \tan^2 A = k(1 - 3\tan^2 A)$

$\Rightarrow \quad 3 - \tan^2 A = k - 3k\tan^2 A$

$\Rightarrow \quad 3k\tan^2 A - \tan^2 A = k - 3$

$\Rightarrow \quad \tan^2 A(3k - 1) = k - 3$

$\Rightarrow \quad \tan^2 A = \dfrac{k-3}{3k-1}$

where $\dfrac{k-3}{3k-1} > 0 \Rightarrow k < \dfrac{1}{3}$ or $k > 3$

Now, $\operatorname{cosec} A\,(3\sin A - 4\sin^3 A)$

$= \operatorname{cosec} A \times \sin A\,(3 - 4\sin^2 A)$

$= \dfrac{1}{\sin A} \times \sin A\,(3 - 4\sin^2 A)$

$\left(\because \operatorname{cosec} A = \dfrac{1}{\sin A}\right)$

$= 3 - 4\sin^2 A = 3 - 4\left[\dfrac{\tan^2 A}{1 + \tan^2 A}\right]$

$= \dfrac{3 - \tan^2 A}{1 + \tan^2 A} = \dfrac{3 - \dfrac{k-3}{3k-1}}{1 + \dfrac{k-3}{3k-1}}$

$= \dfrac{9k - 3 - k + 3}{3k - 1 + k - 3}$

$= \dfrac{8k}{4k - 4} = \dfrac{2k}{k-1},$

where $k < \dfrac{1}{3}$ or $k > 3$

**182.** *(c)* Given, $p = \sqrt{\dfrac{1 - \sin x}{1 + \sin x}}$

$p = \sqrt{\dfrac{(1 - \sin x)(1 - \sin x)}{(1 + \sin x)(1 - \sin x)}}$

$= \dfrac{1 - \sin x}{\sqrt{1 - \sin^2 x}} = \dfrac{1 - \sin x}{\cos x}$

$r = \dfrac{\cos x}{1 + \sin x} \times \dfrac{(1 - \sin x)}{(1 - \sin x)} = \dfrac{\cos(1 - \sin x)}{1 - \sin^2 x}$

$r = \dfrac{\cos x\,(1 - \sin x)}{\cos^2 x} = \dfrac{1 - \sin x}{\cos x}$

$\Rightarrow \quad p = q = r$

Also, $q \times r = \dfrac{1 - \sin x}{\cos x} \times \dfrac{\cos x}{1 + \sin x}$

$= \dfrac{1 - \sin x}{1 + \sin x} = p^2$

Hence, both statements are correct.

**183.** *(a)* I. LHS $= \dfrac{\cos A}{1 - \tan A} + \dfrac{\sin A}{1 - \cot A}$

$= \dfrac{\cos A}{1 - \dfrac{\sin A}{\cos A}} + \dfrac{\sin A}{1 - \dfrac{\cos A}{\sin A}}$

$= \dfrac{\cos^2 A}{\cos A - \sin A} + \dfrac{\sin^2 A}{\sin A - \cos A}$

$= \dfrac{\cos^2 A - \sin^2 A}{(\cos A - \sin A)}$

$= \dfrac{(\cos A - \sin A)(\cos A + \sin A)}{(\cos A - \sin A)}$

$= (\cos A + \sin A) =$ RHS

II. $[(1 - \sin A) - \cos A]^2$

$= (1 - \sin A)^2 + \cos^2 A - 2\cos A(1 - \sin A)$

$= 1 + \sin^2 A - 2\sin A + \cos^2 A - 2\cos A(1 - \sin A)$

$= 1 + \sin^2 A + \cos^2 A - 2\sin A - 2\cos A(1 - \sin A)$

$= 2 - 2\sin A - 2\cos A(1 - \sin A)$

$= 2(1 - \sin A) - 2\cos A(1 - \sin A)$

$= 2(1 - \sin A)(1 - \cos A) \neq$ RHS

Hence, only statement I is correct.

**184.** *(a)* We have, $\sin\theta = \dfrac{2a + 3b}{3b} = 1 + \dfrac{2a}{3b}$

Since, $\sin\theta$ is always smaller or equal to 1 but $1 + \dfrac{2a}{3b} > 1$.

Hence, it is not possible.

**185.** *(a)* Since, $0 \le \cos^2 x \le 1$

$\because \; -1 \le \cos^2 x + \cos^2 y - \cos^2 z \le 2$

$\therefore$ Minimum value of the given expression is $-1$.

**186.** *(a)* We have, $\tan\theta + \sec\theta = 2$

$\Rightarrow \quad \sec\theta = 2 - \tan\theta$

On squaring both sides, we get

$$\sec^2\theta = 4 + \tan^2\theta - 4\tan\theta$$

$\Rightarrow \quad 1 + \tan^2\theta = 4 + \tan^2\theta - 4\tan\theta$

$\Rightarrow \quad 4\tan\theta = 3 \Rightarrow \tan\theta = \dfrac{3}{4}$

**187.** *(b)* We have, $\dfrac{\cos\theta}{1 - \sin\theta} \times \dfrac{1 + \sin\theta}{1 + \sin\theta}$

$= \dfrac{\cos\theta(1 + \sin\theta)}{1 - \sin^2\theta} = \dfrac{\cos\theta(1 + \sin\theta)}{\cos^2\theta}$

$= \dfrac{1 + \sin\theta}{\cos\theta}$

**188.** *(c)* We have,

$\tan(x + 40)° \tan(x + 20)° \tan(3x)° \tan(70 - x)° \tan(50 - x)° = 1$

$[\because \; \cot(90° - \theta) = \tan\theta]$

$\Rightarrow \quad \tan(x + 40)° \tan(x + 20)° \tan(3x)° \cot(90° - 70° + x)° \cot(90° - 50° + x)° = 1$

$\Rightarrow \quad \tan(x + 40)° \tan(x + 20)° \tan(3x)° \cot(20 + x)° \cot(40 + x)° = 1$

$\Rightarrow \quad \tan(3x)° = 1 \Rightarrow \tan(3x)° = \tan 45°$

$\Rightarrow \quad 3x = 45 \quad \Rightarrow \quad x = 15$

**189.** *(d)* We have, $32\cot^2\dfrac{\pi}{4} - 8\sec^2\left(\dfrac{\pi}{3}\right) + 8\cos^3\left(\dfrac{\pi}{6}\right)$

$= 32 \cdot (1) - 8 \cdot (2)^2 + 8 \cdot \left(\dfrac{\sqrt{3}}{2}\right)^3$

$= 32 - 8 \cdot (4) + 8 \cdot \dfrac{3\sqrt{3}}{8}$

$= 32 - 32 + 3\sqrt{3} = 3\sqrt{3}$

**190.** *(b)* We have, $x = a\cos\theta$ and $y = b\cot\theta$

$\therefore \left(\frac{a}{x}-\frac{b}{y}\right)\left(\frac{a}{x}+\frac{b}{y}\right)=\frac{a^2}{x^2}-\frac{b^2}{y^2}$

$= \sec^2\theta - \tan^2\theta = 1$

**191.** *(a)* We have, $\sin\theta\cos\theta = 2\cos^3\theta - \frac{3}{2}\cos\theta$

$\Rightarrow \quad 2\sin\theta = 4\cos^2\theta - 3$

$\Rightarrow \quad 2\sin\theta = 4 - 4\sin^2\theta - 3$

$\Rightarrow \quad 4\sin^2\theta + 2\sin\theta - 1 = 0,$

$\Rightarrow \quad \sin\theta = \frac{-2\pm\sqrt{4+16}}{8} = \frac{-2\pm 2\sqrt{5}}{8}$

$\Rightarrow \quad \sin\theta = \frac{-1\pm\sqrt{5}}{4}$

Since, $\theta$ is acute angle

$\therefore \quad \sin\theta > 0$

$\Rightarrow \quad \sin\theta = \frac{\sqrt{5}-1}{4}$

**192.** *(d)* Hours moved by hour hand = 5 hrs 10 min

$= 5 + \frac{10}{60} = \frac{31}{6}$ hrs

Angle traced by hour hand in 1 hr = 30°

$\therefore$ Angle traced by hour hand is

$\left(\frac{31}{6}\right)$ hrs $= 30\times\frac{31}{6} = 155°$

**193.** *(b)* I. If $45° < \theta < 90°$, then $\sin\theta > \cos\theta$

$\therefore \quad \sin 66° > \cos 66°$

II. When $0° < \theta < 45°$, $\cos\theta > \sin\theta$

$\therefore \quad \cos 26° > \sin 26°$

or $\quad \sin 26° < \cos 26°$.

Hence only II is correct.

**194.** *(a)* I. LHS $= \frac{1+\tan^2\theta}{1+\cot^2\theta} = \frac{\sec^2\theta}{\text{cosec}^2\theta}$

$= \tan^2\theta$

RHS $= \left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2 = \left[\frac{\tan\theta[\cot\theta-1]}{1-\cot\theta}\right]^2$

$= (-\tan\theta)^2 = \tan^2\theta$

$\therefore$ LHS = RHS

II. $\cot\theta = \frac{1}{\tan\theta} \Rightarrow \tan\theta\cdot\cot\theta = 1$

which true for all value of $\theta$.

So, statement II is false.

Hence, only statement I is correct.

**195.** *(d)* We have, $A + B + C = \pi$

$\Rightarrow \quad \frac{\pi}{2} + B + B + C = \pi\ [A = B + \pi/2]$

$\Rightarrow \quad 2B + C = \pi - \frac{\pi}{2} = \frac{\pi}{2}$

**196.** *(c)* $\left(\frac{\sin 35°}{\cos 55°}\right)^2 - \left(\frac{\cos 55°}{\sin 35°}\right)^2 + 2\sin 30°$

$= \left[\frac{\sin(90°-55°)}{\cos 55°}\right]^2 - \left[\frac{\cos(90°-35°)}{\sin 35°}\right]^2 + 2\times\frac{1}{2}$

$= \left(\frac{\cos 55°}{\cos 55°}\right)^2 - \left(\frac{\sin 35°}{\sin 35°}\right)^2 + 1$

$= 1^2 - 1^2 + 1 = 1$

**197.** *(c)* We have, $\tan\theta + \cot\theta = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{\sin^2\theta + \cos^2\theta}{\cos\theta\cdot\sin\theta} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{1}{\frac{\sin 2\theta}{2}} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \sin 2\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$

$\therefore \quad \sin\theta + \cos\theta = \sin\frac{\pi}{6} + \cos\frac{\pi}{6}$

$= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}+1}{2}$

**198.** *(b)* We have,

$p = \cot\theta + \tan\theta = \text{cosec}\theta\cdot\sec\theta$

$q = \sec\theta - \cos\theta = \sin^2\theta\cdot\sec\theta$

$\therefore \quad (p^2q)^{2/3} - (q^2p)^{2/3}$

$= (\text{cosec}^2\theta\cdot\sec^2\theta\cdot\sin^2\theta\cdot\sec\theta)^{2/3}$

$- (\sin^4\theta\cdot\sec^2\theta\cdot\text{cosec}\theta\cdot\sec\theta)^{2/3}$

$= (\sec^3\theta)^{2/3} - (\sin^3\theta\cdot\sec^3\theta)^{2/3}$

$= \sec^2\theta - \sin^2\theta\cdot\sec^2\theta$

$= \sec^2\theta(1-\sin^2\theta) = 1$

**199.** *(d)* Given, $\frac{x}{a} - \frac{y}{b}\tan\theta = 1 \quad$ ...(i)

and $\quad \frac{x}{a}\tan\theta + \frac{y}{b} = 1 \quad$ ...(ii)

On solving Eqs. (i) and (ii), we get

$\frac{x}{a} = \frac{1+\tan\theta}{\sec^2\theta}$ and $\frac{y}{b} = \frac{1-\tan\theta}{\sec^2\theta}$

$\therefore \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{(1+\tan\theta)^2 + (1-\tan\theta)^2}{\sec^4\theta}$

$= \frac{2\sec^2\theta}{\sec^4\theta}$

$= 2\cos^2\theta$

**200.** *(c)* $\because \quad 3 - \tan^2\theta = \alpha(1 - 3\tan^2\theta)$

$\Rightarrow \quad (3\alpha - 1)\tan^2\theta = \alpha - 3$

$\Rightarrow \quad \tan^2\theta = \frac{\alpha-3}{3\alpha-1}$

As, $\tan^2\theta \geq 0$, then $\frac{\alpha-3}{3\alpha-1} \geq 0 \Rightarrow \alpha \geq 3$

or, $\alpha < \frac{1}{3} \quad \Rightarrow \quad \alpha\in\left(-\infty, \frac{1}{3}\right)\cup[3,\infty)$

**201.** *(a)* To exchange the position, both hands has to cover 360° together.

Angle traced by hour hand in 1 min

$= (1/2)^\circ$

Angle traced by minute hand in 1 min = 6°

Let the required time be $t$ min. Then,

$6t + \frac{1}{2}t = 360° \Rightarrow t = \frac{360\times 2}{13}$

$= \frac{720}{13}$ min = 55.38 min

**202.** *(c)* I. $\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}\times\frac{1-\cos\theta}{1-\cos\theta}}$

$= \sqrt{\frac{(1-\cos\theta)^2}{1-\cos^2\theta}} = \sqrt{\frac{(1-\cos\theta)^2}{\sin^2\theta}}$

$= \frac{1-\cos\theta}{\sin\theta} = \text{cosec}\,\theta - \cot\theta$

Hence, statement I is true.

II. $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \sqrt{\frac{1+\cos\theta}{1-\cos\theta}\times\frac{1+\cos\theta}{1+\cos\theta}}$

$= \sqrt{\frac{(1+\cos\theta)^2}{\sin^2\theta}} = \frac{1+\cos\theta}{\sin\theta}$

$= \text{cosec}\,\theta + \cot\theta$

Hence, statement II is true.

**203.** *(b)* $\because \frac{\cos^2\theta - 3\cos\theta + 2}{\sin^2\theta} = 1$

$\Rightarrow \quad \cos^2\theta - 3\cos\theta + 1 + 1 - \sin^2\theta = 0$

$\Rightarrow \quad \cos^2\theta - 3\cos\theta + 1 + \cos^2\theta = 0$

$\Rightarrow \quad 2\cos^2\theta - 3\cos\theta + 1 = 0$

$\Rightarrow \quad (2\cos\theta - 1)(\cos\theta - 1) = 0$

$\Rightarrow \quad \cos\theta = \frac{1}{2}$ or, $\cos\theta = 1$

$\Rightarrow \quad \theta = \frac{\pi}{3} [\because\ 0 < \theta < \pi/2]$

Hence, only II is correct.

**204.** *(d)* I. As, $\cos x$ lies between −1 and 1, then $\cos x = 2^{m+1}$ does not exist for positive value of $m$.

II. The given relation $mn \geq m + n$ is true for $m, n \in N$ and $m > 1$ and $n > 1$.

Thus, statement II is not true.

Hence, neither I nor II is correct

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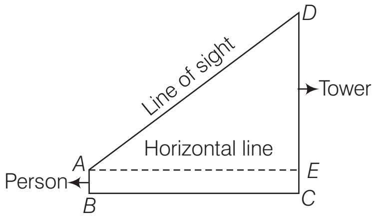
# HEIGHT AND DISTANCE

*Regularly (1-2) questions have been asked from this chapter. Generally this is the simplest application of trigonometry related to our real world and a little practice can get you command over this section.*

It is an important application of trigonometry which helps us to find the height of any object and distance of that object from any point which are not directly measurable. If the angle of elevation/depression from a point is known.

### Line of sight

A line of sight is the line drawn from the eye of an observer to the point, where the object is viewed by the observer.



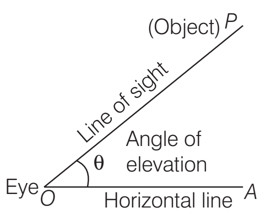

### Horizontal Line

The line of sight which is parallel to ground level is known as horizontal line.

## ANGLE OF ELEVATION

Angle of elevation is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the upward direction.

Let $P$ be the position of the object above the horizontal line $OA$ and $O$ be the eye of the observer, then $\angle AOP$ is called angle of elevation.




**IMPORTANT POINTS**

- If the observer moves towards the perpendicular line (tower/building), then angle of elevation increases and if the observer moves away from the perpendicular line (tower/building), then the angle of elevation decreases.
- If the angle of elevation of Sun above a tower decreases, then the length of shadow of a tower increases.
- If height of tower is doubled and the distance between the observer and foot of the tower is also doubled, then the angle of elevation remains same.

**EXAMPLE 1.** The angle of elevation of the top of a tower of height $100\sqrt{3}$ m from a point at a distance of 100 m from the foot of the tower on a horizontal plane is

**a.** 30° **b.** 60° **c.** 75° **d.** 105°

***Sol.*** **b.** Let $\phi$ be the angle of elevation.

Given, $AC$ = Height of tower

$= 100\sqrt{3}$ m

and $AB = 100$ m

In right angled $\Delta ABC$,

$$\tan\phi = \frac{AC}{AB} = \frac{100\sqrt{3}}{100} = \sqrt{3}$$

$\Rightarrow \quad \tan\phi = \tan 60°$

$\Rightarrow \quad \phi = 60°$

Hence, angle of elevation is 60°.